

Homework 1: Surface Pressure, Lift, and Pressure Drag

NACA 2412 at $\alpha = 5^\circ$ | Revised September 28, 2026

Name: _____

UMass ID: _____

Due: see Canvas

Section: _____

Purpose and model

Compute the two-dimensional pressure force on a NACA 2412 airfoil. Question 1 uses continuous geometry and pressure functions. Question 2 uses 101 supplied stations to compute the same force by summing straight-panel contributions. No derivation of the pressure-force formulas is required.

Model statement. The supplied pressure distribution is an instructional surrogate for attached flow at $\alpha = 5^\circ$. It gives a plausible suction peak and lift trend, but it is not measured pressure-tap data or a universal NACA 2412 pressure law. Only pressure forces are included: the computed drag is pressure drag, not total drag.

Given conditions and conventions

Quantity	Value	Quantity	Value
Airfoil	NACA 2412	Angle of attack	$\alpha = 5.0^\circ$
Chord	$c = 0.500$ m	Reference span	$b = 1.000$ m
Freestream speed	$V_\infty = 30.0$ m/s	Density	$\rho_\infty = 1.225$ kg/m ³
Static pressure	$p_\infty = 101325$ Pa	Dynamic pressure	$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$

Write $X = x/c$ and $Y = y/c$. Positive x points from leading edge to trailing edge; positive y points upward. Subscripts u and l denote upper and lower surfaces.

$$C_p = \frac{p - p_\infty}{q_\infty}$$

Positive C_N points upward normal to the chord. Positive C_A points along the chord from leading edge to trailing edge. Rotate these coefficients into wind axes:

$$C_L = C_N \cos \alpha - C_A \sin \alpha$$

$$C_{D,p} = C_N \sin \alpha + C_A \cos \alpha$$

NACA 2412 geometry for both questions

Use $m = 0.02$, $p = 0.40$ and $t = 0.12$. The following common-x approximation supplies upper and lower ordinates.

$$Y_c(X) = \begin{cases} \frac{m}{p^2}(2pX - X^2) & 0 \leq X \leq p \\ \frac{m}{(1-p)^2}[1 - 2p + 2pX - X^2] & p < X \leq 1 \end{cases}$$

$$Y_t(X) = 5t(0.2969\sqrt{X} - 0.1260X - 0.3516X^2 + 0.2843X^3 - 0.1036X^4)$$

$$Y_u = Y_c + Y_t$$

$$Y_l = Y_c - Y_t$$

The derivatives below are supplied for numerical integration. Students need not derive them.

$$\frac{dY_c}{dX} = \begin{cases} \frac{2m}{p^2}(p - X) & 0 \leq X \leq p \\ \frac{2m}{(1-p)^2}(p - X) & p < X \leq 1 \end{cases}$$

$$\frac{dY_t}{dX} = 5t\left(\frac{0.2969}{2\sqrt{X}} - 0.1260 - 0.7032X + 0.8529X^2 - 0.4144X^3\right)$$

Therefore $dY_u/dX = dY_c/dX + dY_t/dX$ and $dY_l/dX = dY_c/dX - dY_t/dX$.

Question 1 - Continuous surface integration [45 points]

Use this smooth pressure-loading model for $0 \leq X \leq 1$:

$$C_0(X) = (1 - X)[e^{-X/0.010} + 0.20(1 - e^{-X/0.020})]$$

$$\Delta C_p(X) = 0.721896 \frac{(1 - e^{-X/0.010})(1 - X)}{\sqrt{X + 0.020}}$$

$$C_{p,u} = C_0 - 0.80\Delta C_p$$

$$C_{p,l} = C_0 + 0.20\Delta C_p$$

Use $p_u = p_\infty + q_\infty C_{p,u}$ and $p_l = p_\infty + q_\infty C_{p,l}$ if dimensional pressures are needed.

1a. Decode the four digits of NACA 2412. Calculate q_∞ and state its units.

1b. Plot Y_u and Y_l on equal axes. Separately plot $C_{p,u}$ and $C_{p,l}$ versus X with the conventional inverted C_p axis. Label the leading-edge suction region and the pressure difference that produces positive lift.

1c. Numerically evaluate the supplied pressure-force integrals for C_N and C_A . Do not derive these formulas. Follow the numerical-integration guide below and document your method and convergence.

$$C_N = \int_0^1 (C_{p,l} - C_{p,u}) dX$$

$$C_A = \int_0^1 [C_{p,u} \frac{dY_u}{dX} - C_{p,l} \frac{dY_l}{dX}] dX$$

Numerical-integration guide for 1c

Why the leading edge needs care. The thickness slope contains $1/\sqrt{X}$, which is undefined at $X = 0$. The area under it is finite, but a numerical routine must not evaluate that raw slope at $X = 0$.

Recommended route. Set $X = s^2$, so $dX = 2s ds$ and $0 \leq s \leq 1$. This places more X -points near the leading edge and removes the endpoint difficulty. Define the transformed normal and axial integrands as:

$$g_N(s) = 2s[C_{p,l}(s^2) - C_{p,u}(s^2)]$$

$$g_A(s) = 2s[C_{p,u}(s^2)Y_u'(s^2) - C_{p,l}(s^2)Y_l'(s^2)]$$

Here a prime means dY/dX , evaluated at $X = s^2$. At the single endpoint $s = 0$, use the finite limits $g_N(0) = 0$ and $g_A(0) = 10t(0.2969) = 0.35628$; do not calculate $0 \times \infty$.

Compute and check. Use the composite trapezoidal rule for each g on $s_j = j/N$, $j = 0, \dots, N$:

$$I[g] \approx \frac{1}{N} \left(\frac{1}{2} g(0) + \sum_{j=1}^{N-1} g\left(\frac{j}{N}\right) + \frac{1}{2} g(1) \right)$$

Here $I[g]$ means the integral of g from $s = 0$ to 1. Apply the rule once with $N = 1000$ and again with $N = 2000$. Compare the two C_N values and the two C_A values. Report both discretizations and their absolute differences. A difference below 10^{-4} for each coefficient is adequate for this assignment.

What is adaptive quadrature? It is software that estimates integration error and automatically uses smaller intervals where a function changes rapidly. You are not asked to implement it. If you choose a library such as `scipy.integrate.quad`, apply it to g_N and g_A on $0 \leq s \leq 1$, state the requested tolerance, and repeat with a tighter tolerance to demonstrate convergence.

1d. Use C_N and C_A to calculate C_L , $C_{D,p}$, and the forces per unit span. Report coefficients to four significant figures, forces with units, and do not call pressure drag total drag.

$$L' = q_\infty c C_L$$

$$D'_p = q_\infty c C_{D,p}$$

Question 2 - Tabulated data and panel sums [45 points]

Use the companion workbook **MIE446_HW1_NACA2412_Data.xlsx**. The **Data** sheet supplies 101 stations (X , Y_u , Y_l , p_u , p_l) from $X = 0$ to 1. Preserve the supplied data columns. Consecutive stations define 100 straight panels; panel i connects stations i and $i + 1$.

2a. Add columns for $C_{p,u,i}$ and $C_{p,l,i}$ calculated from the dimensional pressures and the given p_∞ and q_∞ .

2b. For each of the 100 panels, calculate the mean upper and lower C_p values, the changes in X , Y_u , and Y_l , and the two force contributions below. Show one row per panel and keep the formulas visible. No finite-difference surface slopes are needed.

$$\bar{C}_{p,u,i} = \frac{C_{p,u,i} + C_{p,u,i+1}}{2} \quad \Delta X_i = X_{i+1} - X_i$$

Use the same mean formula for the lower surface. The force contributions are:

$$\Delta C_{N,i} = \frac{(C_{p,l,i} - C_{p,u,i}) + (C_{p,l,i+1} - C_{p,u,i+1})}{2} \Delta X_i$$

$$\Delta C_{A,i} = \bar{C}_{p,u,i}(Y_{u,i+1} - Y_{u,i}) - \bar{C}_{p,l,i}(Y_{l,i+1} - Y_{l,i})$$

2c. Sum the 100 contributions for C_N and C_A . Rotate them into C_L and $C_{D,p}$, then calculate L' and D_p' . Compare C_L and $C_{D,p}$ with Question 1 using the percentage-difference formula below.

$$C_N \approx \sum_{i=1}^{100} \Delta C_{N,i} \quad C_A \approx \sum_{i=1}^{100} \Delta C_{A,i}$$
$$\text{Percentage difference (\%)} = 100 \times \frac{|\text{panel result} - \text{continuous result}|}{|\text{continuous result}|}$$

2d. In two or three sentences, compare the lift and pressure-drag differences. Pressure drag can be more sensitive to panel resolution because its two wind-axis terms partly cancel. Do not infer skin-friction drag from the pressure data.

Submission and grading [10 points for communication]

Submit through Canvas: (1) one PDF with plots, results, units, numerical-convergence table, and the Question 1-Question 2 comparison; (2) the completed workbook with original data intact and panel formulas visible; (3) one Claim-Evidence-Check summary slide in PPTX and PDF; and (4) a brief AI-use disclosure. Show your own reasoning. AI may help with syntax or debugging, but it cannot substitute for independent verification.

Criterion	Points
Q1: NACA and pressure interpretation; correctly labeled plots	18
Q1: supplied integrals, documented convergence, coefficients and forces	27
Q2: pressure coefficients and traceable panel contributions	20
Q2: panel sums, rotation, comparison and interpretation	25
Readable work, units, summary slide and AI disclosure	10
Total	100

References

E. N. Jacobs, K. E. Ward and R. M. Pinkerton, *The Characteristics of 78 Related Airfoil Sections from Tests in the Variable-Density Wind Tunnel*, NACA Report 460, [NASA NTRS](#). NASA OpenVSP Ground School, [NACA 4-Series Airfoil](#). NASA Glenn Research Center, [Lift Equation](#).